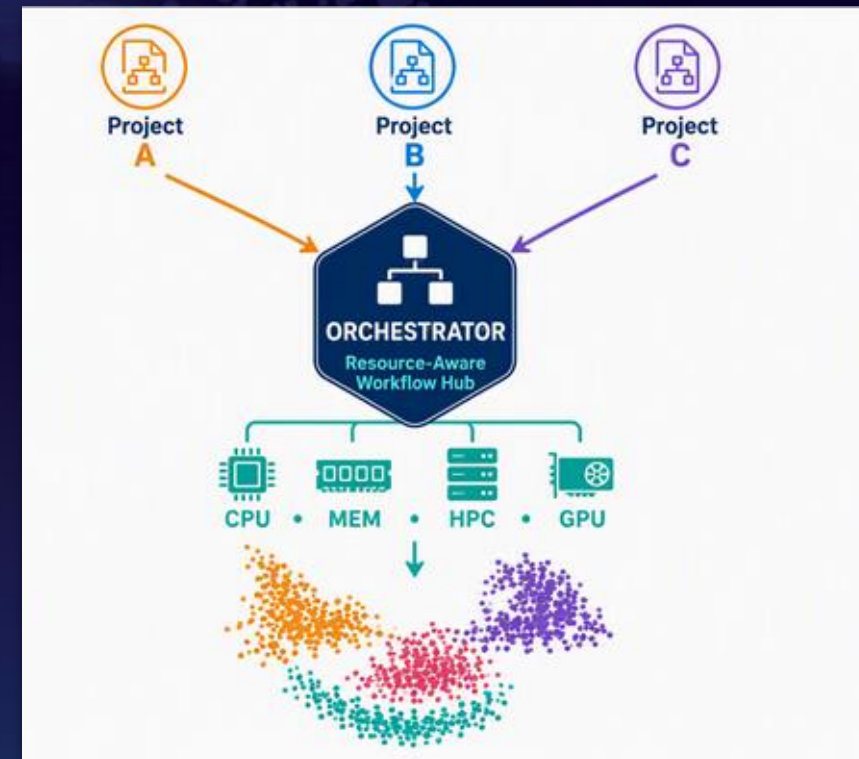


Resource-Aware Multi-Project Workflow Orchestrator for Scalable Single-Cell Analysis

Daedalus

Smarter Resources.
Faster Workflows.
Scalable Science.

Team 15



Team 15

- **Antonia Chroni.** Project Lead. Lead Bioinformatics Research Scientist, DNB, St. Jude Children's Research Hospital.
- **Emma Bishop.** Data Scientist, Fred Hutch Cancer Center.
- **A. S. M. Tanjim Hassan.** Undergraduate Student, University of Dhaka.
- **Sarthak Sunil Dhanke.** Research Assistant, University of Chicago.
- **Clay McLeod.** Director of Product Development and Engineering, St. Jude Children's Research Hospital.
- **Rachana Pandey.** Graduate Research Assistant, University of Minnesota, Twin Cities.
- **Rojina Sapkota.** Bioinformatics Analyst, CPNDR, St. Jude Children's Research Hospital.
- **Lindsey Warren.** Enterprise Architect, International Paper.



Problem: Manual LSF wiring slows single-cell analysis

- Today, bioinformaticists run **Automated Single-cell Pipelines** on St Jude HPC using a system called **SC-RNA-SEQ-SNAP**.
- It requests a fixed amount resources (CPU, RAM).
- There is no pre-submission validation — errors surface only after jobs are queued and fail on the HPC.
- Single-cell analysis projects can require:
 - larger cohorts (>70 samples or >500,000 cells)
 - concurrent execution of multiple projects (e.g., 15+ projects simultaneously)



Solution & What We Built: Daedalus

Smarter Resources. Faster Workflows. Scalable Science.

⚙️ *Pre-existing: Single YAML configuration, containerized execution, and automated parallel job submission*

- ✓ Integrated SNAP + WDL + Sprocket + HPC
- ✓ Enabled dependency-aware parallel execution
- ✓ Tested two core workflow modules
- ✓ Automated documentation and reporting (GitHub pages)
- ✓ Established a scalable foundation for multi-project workflows and future pipeline expansion.



Snap pipeline, Victoria Knockout cohort (4 samples · ~27k cells) - Honnell et al 2022. Nature Communications.



What changes from Snap Legacy to Daedalus?



Snap Legacy script

- **One YAML Workflow Profile**
containerized execution
- **Automated parallel job submission**
- **Static, global resource setting**
Defaults are designed for a boilerplate project; optimization per cohort is required.
- **Manual LSF wiring**
Requires separate `bsub` (workload scheduling) calls, bash scripts, and configuration steps.
- **Reactive debugging**
Limited observability. Issues only surface after a failed run.

`launch_full_pipeline.sh` provides a single-command workflow to run all pipeline, parallelization, and toggle modules



Snap Daedalus orchestrator

- **One YAML Workflow Profile**
containerized execution
- **Automated parallel job submission**
- **Dynamic, Cell Ranger-Informed Scaling**
Creates right-sized requests for CPU/RAM based on sample and cell-count metrics
- **WDL + Sprocket**
Sprocket offers dev tools to check and validate LSF before submitting to HPC.
- **launch-snap-downstream**
Provides a single-command workflow to run all pipeline, parallelization, and toggle modules +

Analyst generates WDL from YAML, validates resources, and submits the job to the HPC Cluster.



SNAP-Daedalus Reproduces SNAP-Legacy QC and Integrated Cell Structure of Victoria Knockout

Snap Legacy

sample_name	cells_number_raw	cells_number_seurat_qc	cells_number_doublets	cells_number_final
DYE_2519	4865	3464	NA	3464
DYE_2520	7477	5432	NA	5432
DYE_2521	6351	5778	NA	5278
DYE_2522	8208	7049	NA	7049

 **4865**
raw cells

 **3464**
seurat qc

 **NA**
doublets

 **3464**
final cells

Snap Daedalus

sample_name	cells_number_raw	cells_number_seurat_qc	cells_number_doublets	cells_number_final
DYE_2519	4865	3464	NA	3464
DYE_2520	7477	5432	NA	5432
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 **4865**
raw cells

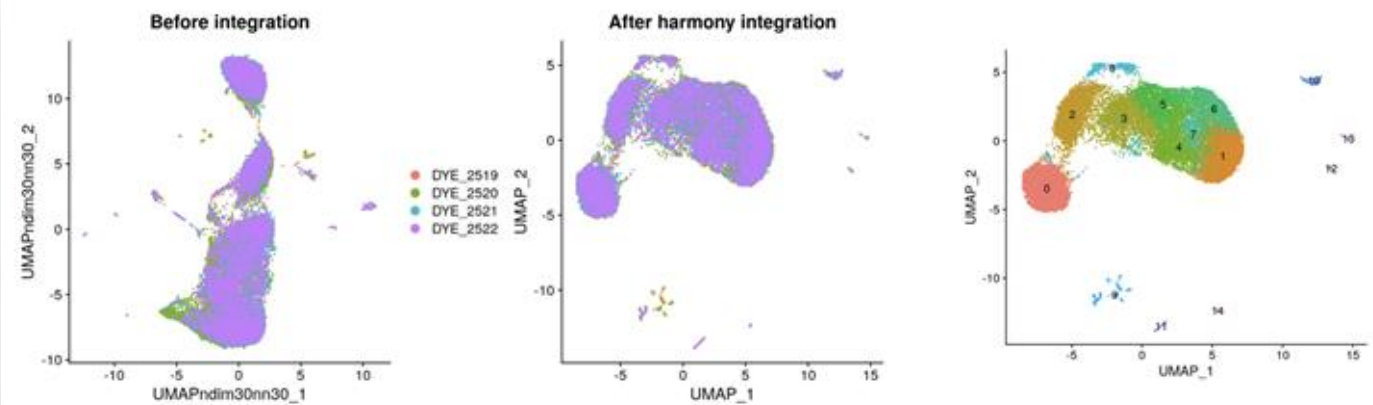
 **3464**
seurat qc

 **NA**
doublets

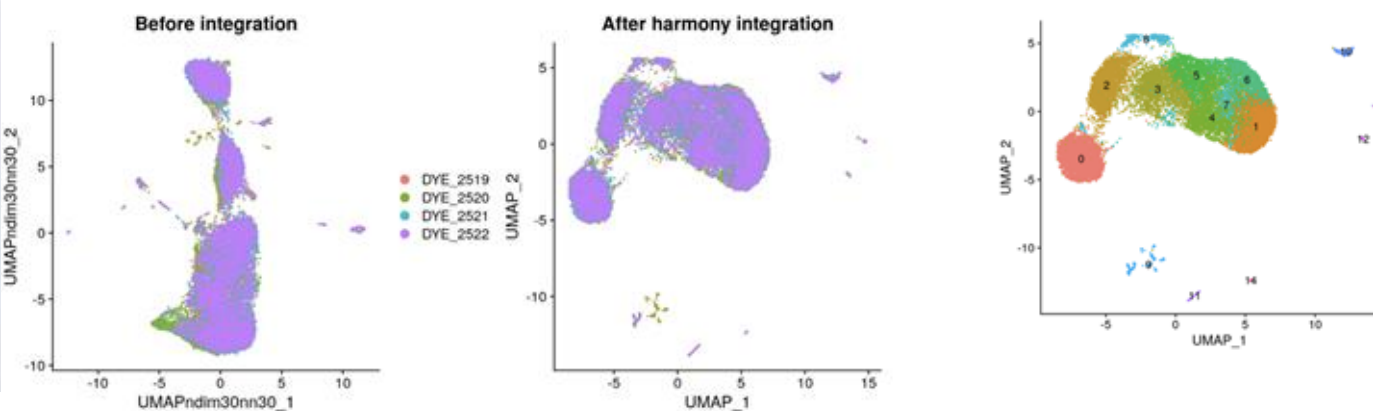
 **3464**
final cells

REPRODUCED

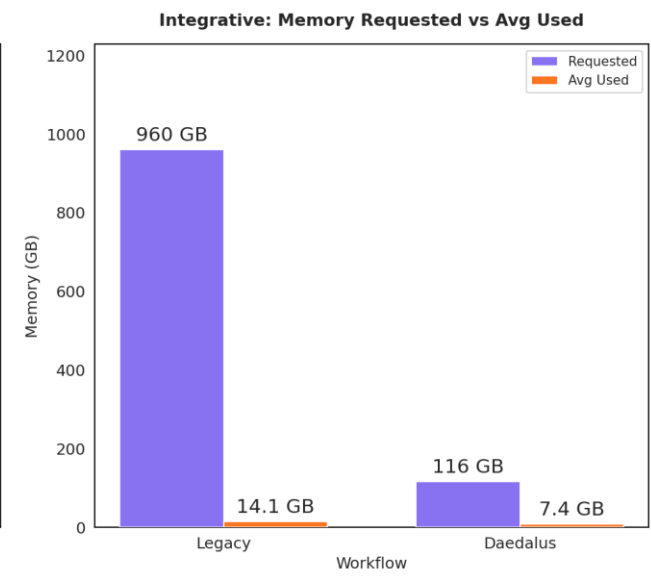
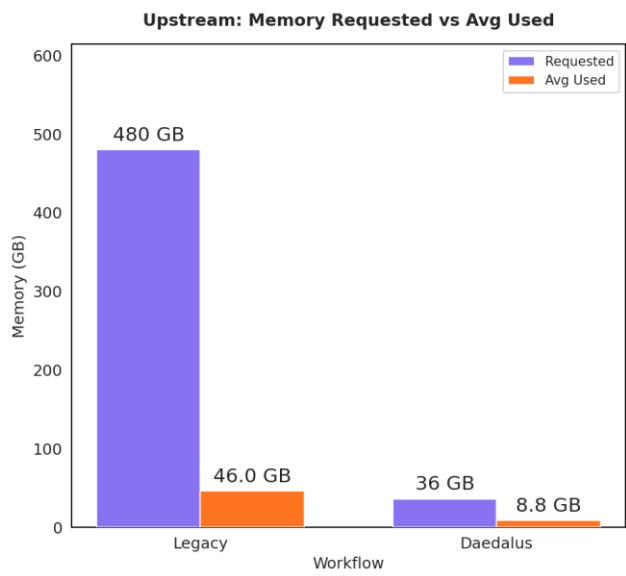
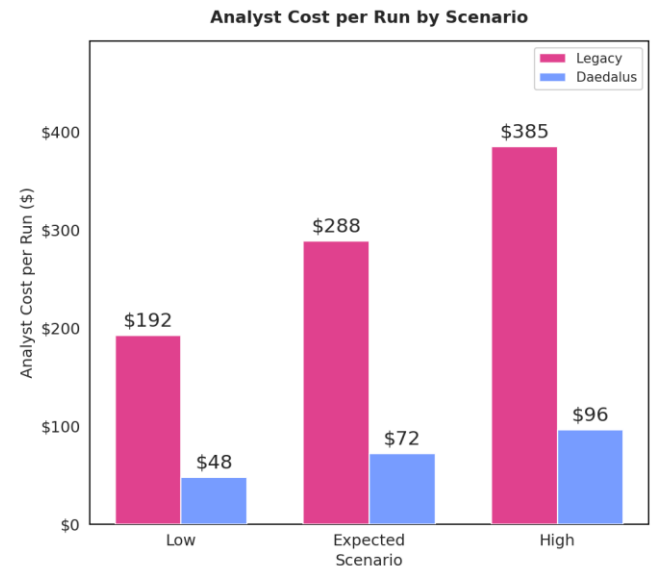
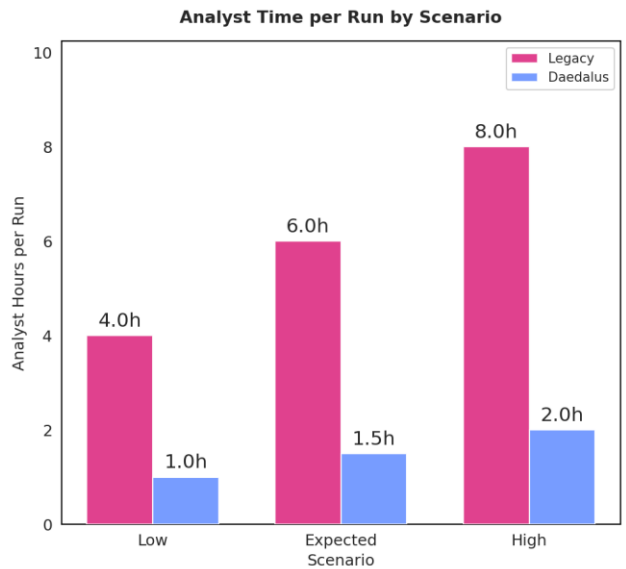
Snap Legacy



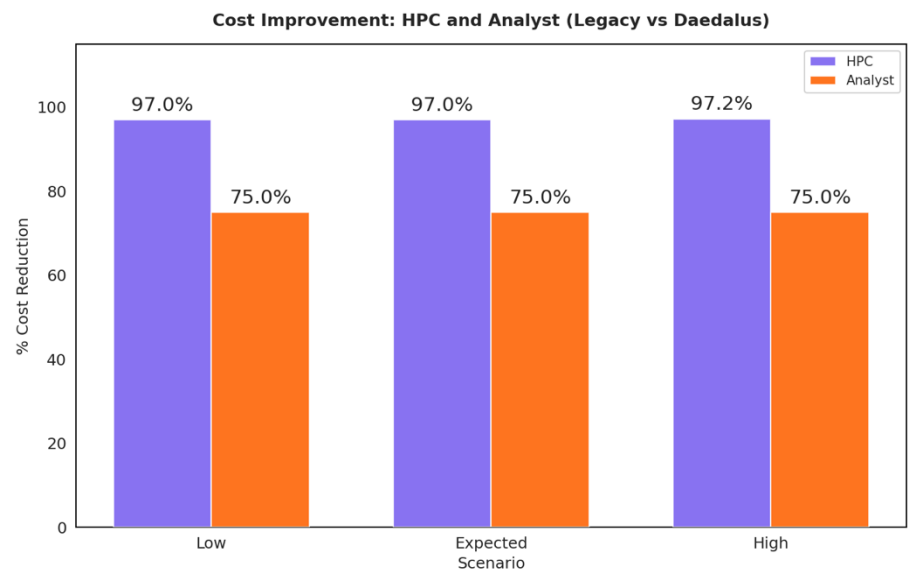
Snap Daedalus



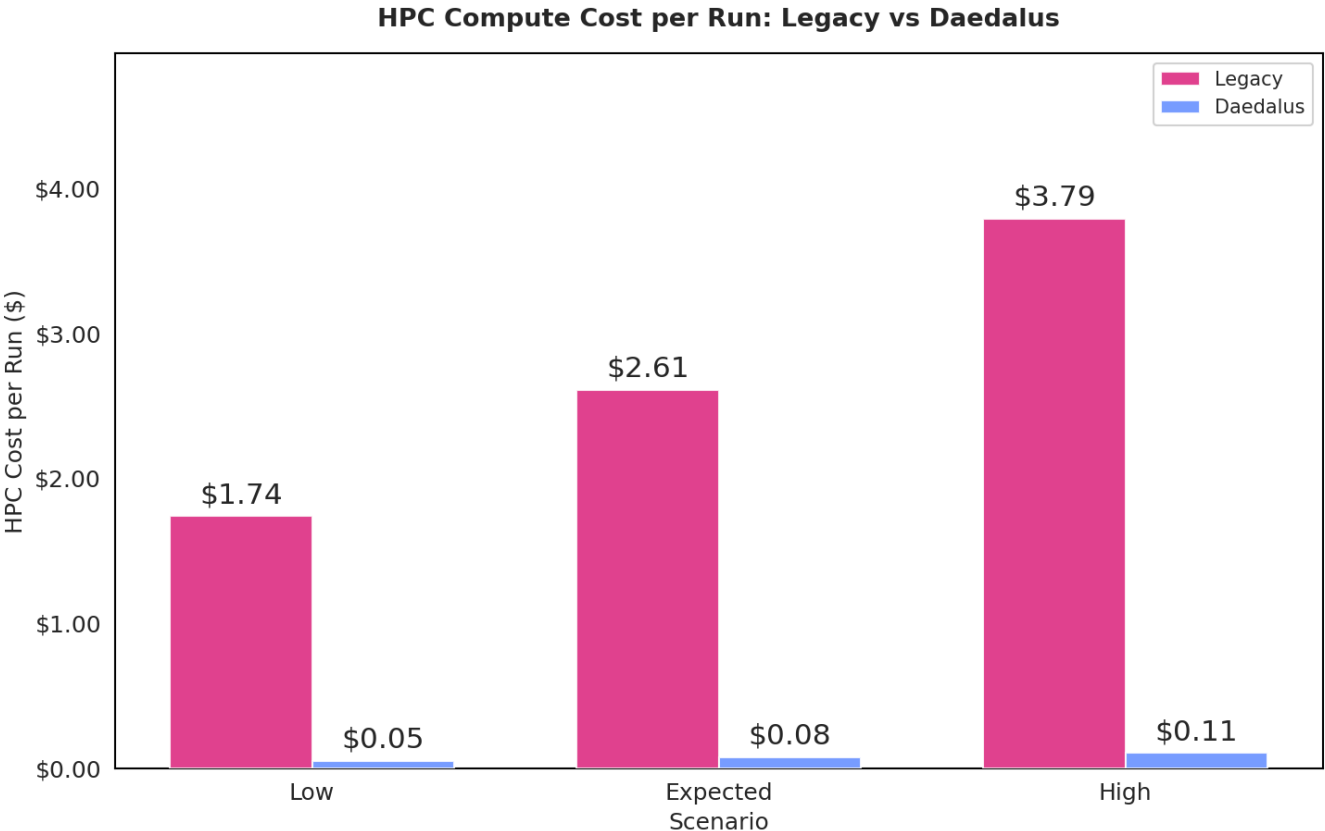
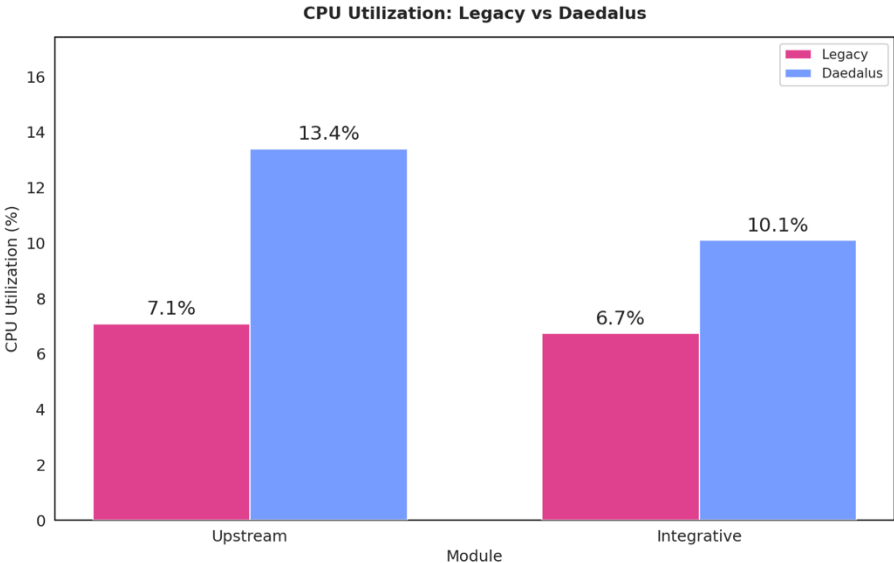
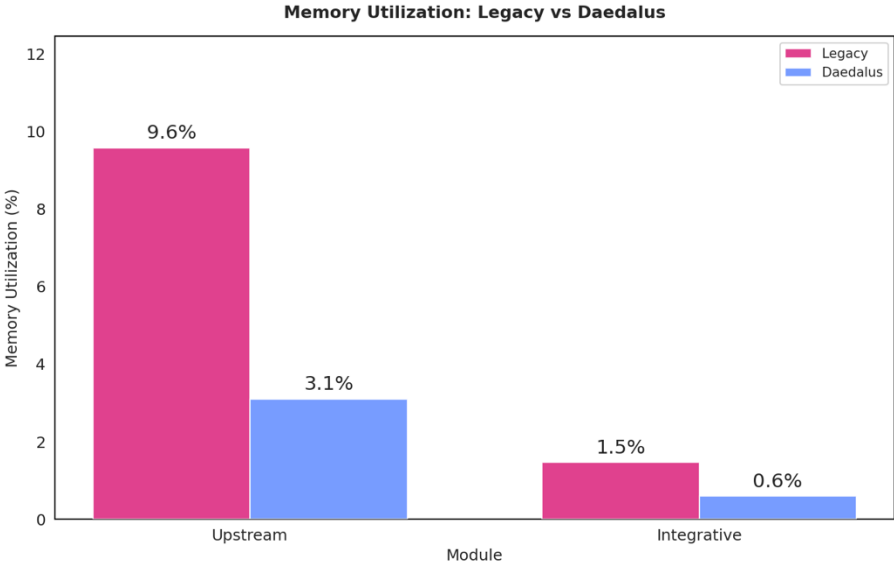
Results and Impact: ROI



- Smarter resource requests use much lower memory allocation.
- ~75% less analyst time per run.
- From \$288/run to \$72/run
- 48.1% ROI at 24 runs/year, with ~14-run payback.



Results and Impact: ROI



Roadmap For the Path Ahead

Pipeline Refactoring

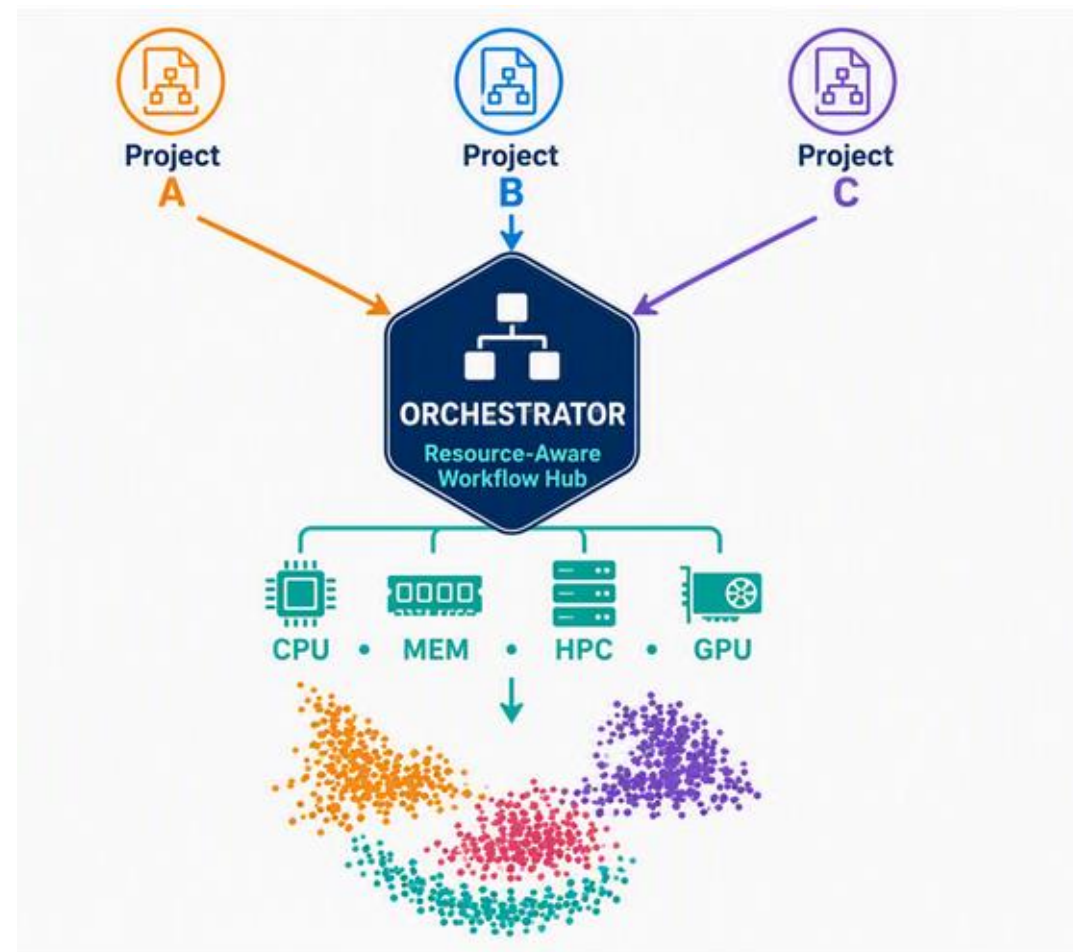
- Define files as **WDL task outputs**
- Enable **call caching** and **unit testing**
- Add metadata for **automated documentation**

Pipeline Expansion

- Extend Daedalus to **scATAC-seq, Parse, and multi-omics**
- Support **multi-project execution** and **adaptive resources**

Collaboration

- Explore [WILDS WDL Writer](#)
- Continue collaboration with St. Jude DNB Bioinformatics core with Computational Biology and HPC team
- Continue collaboration with the **Fred Hutch team**



Thank You

